AgriMind

Comprehensive Crop, Weather & Plant Disease Guide

A practical reference for crop selection, seasonal planning, disease recognition, prevention, integrated management, and weather-aware farming.

Regional focus: Tamil Nadu and comparable Indian farming conditions

Prepared: 14 October 2026 | **Edition:** 2.0 | **Purpose:** Agricultural AI knowledge base

This guide is designed for AgriMind's retrieval-augmented generation (RAG) knowledge base. It provides structured general guidance and links to official agricultural sources. It is not an exhaustive catalogue of every disease or a substitute for current district-level advisories.

Important use and safety notes

Agriculture is highly local. The right crop, planting date, seed variety, irrigation plan, nutrient rate, and disease intervention depend on district, soil, water availability, season, crop stage, local pest pressure, and current official recommendations.

- For crop selection, consult current block/district agrometeorological advisories and TNAU's zone-wise contingency crop plans. A broad seasonal table is a starting point, not a planting order.
- For disease management, identify the likely problem before treatment. Many diseases, insect damage, nutrient deficiencies, herbicide injury, heat stress, and water stress look similar.
- Do not use a pesticide recommendation copied from an old webpage without checking current registration, crop/pest label, dosage, protective equipment, pre-harvest interval, and local advice.
- If disease is severe, fast-spreading, unusual, or high-value crop loss is likely, contact a TNAU/KVK/ICAR extension specialist or plant clinic.

Scope: common examples across rice, maize, millets, pulses, groundnut, cotton, tomato, chilli, onion, mango, banana, and coconut. This is broad coverage, not an exhaustive list.

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1. Weather-aware crop selection and planting decisions

Information needed before choosing a crop

- District, block, field elevation and local agro-climatic zone.
- Rainfall forecast and seasonal outlook; distribution and timing matter, not just the total.
- Water source: reliable irrigation, limited irrigation, or rainfed only.
- Soil texture, drainage, salinity risk, pH, and soil-test results.
- Temperature, heat waves, cold spells, humidity, wind exposure, and expected extreme events.
- Crop duration, sowing window, seed availability, market access, labour, rotation, and disease history.

Broad crop-planning matrix

Use this qualitative matrix to shortlist options for discussion with local extension staff. Local varieties and exact sowing windows differ by zone.

Weather / water situation	Crop groups to evaluate	Key checks before planting
Reliable irrigation; rainfall near normal	Rice in suitable command areas; vegetables, maize, groundnut, pulses, sugarcane or banana where locally suitable.	Soil suitability, water cost, season calendar, drainage, disease history and market.
Rainfed; rainfall reasonably distributed	Millets (pearl millet, finger millet, sorghum), pulses, groundnut, sesame and locally recommended dryland crops.	Sowing after effective rainfall, soil moisture, short/medium-duration varieties and contingency seed.
Deficit rainfall / drought risk	Short-duration millets, pulses, sesame or other drought-adapted options in the local contingency plan.	Current district plan, sowing window, seed availability, crop duration and water budget.
Excess rain / waterlogging risk	Crops and varieties listed in the local excess-rainfall plan; avoid sensitive crops in poorly drained fields.	Drainage, flood exposure, seedbed timing, disease risk and official warning.
Hot, dry or windy conditions	Locally recommended heat/drought-adapted crops and short-duration varieties.	Heat forecast, soil moisture, irrigation capacity and seedling survival.
Cooler / humid periods	Season- and zone-suitable crops with locally recommended varieties.	Fungal/oomycete disease risk, leaf wetness, nursery sanitation, airflow and drainage.

Tamil Nadu reference: TNAU Zone-wise Contingency Crop Plan; it lists crop options by zone and rainfall scenario. Verify the newest plan before planting.

Weather resources: ICAR Weather-based Crop Advisory and IMD Chennai Regional Centre for current district/block advisories.

Seasonal decision checklist

- Do not plant based only on month; verify local season, water release, soil moisture and forecast.
- Use official calendars and crop packages to confirm variety, sowing date, seed rate, spacing and duration.
- Prepare for delayed monsoon, long dry spells, excess rainfall and re-sowing needs.
- For established crops, weather may change irrigation, drainage, scouting and harvest timing even if crop choice cannot change.

2. Weather patterns and crop-health implications

Condition	Risks that may increase	Prevention / response
Warm + humid + long leaf wetness	Some fungal diseases, fruit rots and bacterial diseases; faster spread in dense canopy.	Healthy seed/seedlings, recommended spacing, sanitation, early scouting, drainage and reduced unnecessary leaf wetness.
Frequent rain / waterlogging	Root/collar rots, damping-off, bacterial wilt expression in susceptible crops, nutrient loss.	Clear drainage, avoid moving contaminated soil/water, use healthy transplants and inspect roots/collars.
Dry weather + high temperature	Water stress, scorch, flower/fruit drop, some mite/sucking-pest flare-ups; stress can mimic disease.	Manage soil moisture, mulch where appropriate, irrigate at critical stages and inspect for pests before treatment.
Cloudy / humid weather	Some foliar and seedling diseases; nutrient symptoms can be confused with disease.	Scout frequently, manage infected debris appropriately, maintain spacing/drainage and use crop-specific advice.
Strong wind / storm	Lodging, physical injury, spread of wind/rain-borne spores, spray drift and unsafe operations.	Inspect after storms, avoid spraying in unsafe conditions and follow official warnings.
Dry spell after heavy rain	Alternating water stress and root damage; symptoms can resemble wilt or deficiency.	Check roots, drainage history and field pattern; do not assume all wilting is fungal.

Weather is a risk factor, not proof of disease. AgriMind should combine weather context with crop, stage, symptoms, field distribution and diagnostic evidence.

3. Universal disease prevention and integrated management

Before planting

- Select locally recommended varieties with appropriate resistance/tolerance where available.
- Use reliable disease-free seed and healthy transplants from a reputable nursery.
- Use crop rotation where practical; avoid repeated susceptible hosts when a soil-borne pathogen is known to persist.
- Clean trays, tools and equipment; use clean potting media and avoid contaminated irrigation water.
- Follow locally recommended seed treatment only with currently approved products or validated biological options.

During crop growth

- Scout regularly and record date, crop stage, weather, symptoms, affected area and actions taken.
- Control weeds/volunteer plants where they host pests or pathogens; manage crop residues appropriately.
- Use balanced nutrition; excessive nitrogen and dense growth can worsen some diseases, while deficiencies also weaken crops.
- Maintain crop-appropriate irrigation and drainage. Avoid prolonged leaf wetness where relevant and avoid water stress at sensitive stages.
- Remove severely infected plants or fruit where recommended; clean tools when moving between affected and healthy areas.
- Conserve beneficial insects; use traps or biological control where locally recommended.
- Apply pesticides only when identification and local guidance justify them; follow labels and resistance-management directions.

After harvest

- Manage infected residues according to official guidance; do not move diseased plant material to clean fields.
- Clean tools, containers and storage areas; dry harvested produce appropriately.
- Record disease incidence and interventions to improve next season's crop choice and timing.

4. Crop-wise guide: rice, maize and millets

Rice / paddy

Disease / problem	Typical clues and conditions	Prevention and response
Rice blast	Spindle-shaped leaf lesions with grey centres and dark margins; neck/node lesions can damage panicles. Moist conditions can favour infection.	Use locally recommended varieties and healthy seed; avoid excess nitrogen; monitor nursery and heading stages; confirm suspected neck blast with local experts.
Brown spot	Oval/cylindrical dark-brown spots, sometimes with yellow halo; may affect leaves and glumes.	Healthy seed, balanced nutrition, field sanitation and local confirmation before selecting treatment.
Sheath blight	Oval/irregular green-grey lesions on sheaths near water level, later pale centres and dark margins; dense canopy and humidity can favour spread.	Avoid overly dense planting and excess nitrogen; scout lower sheaths; manage water and residues; follow current local advice.
Bacterial leaf blight	Yellowing or water-soaked streaks from leaf tips/margins; storm injury and wet conditions can aid spread.	Use resistant/tolerant varieties where recommended; avoid plant injury and excess nitrogen; seek extension advice before chemicals.
Sheath rot	Discolouration/lesions on upper sheath around developing panicle; humidity, close planting and injury may contribute.	Healthy seed, suitable spacing, balanced nutrition and monitoring at booting; distinguish from insect injury.
False smut / grain discolouration	Unusual smut balls or discoloured grains near heading/flowering.	Monitor at heading, use current local advisories and follow harvest/storage hygiene.

Sources: TNAU rice blast; brown spot; sheath blight; sheath rot.

Maize

- **Turcicum leaf blight:** elongated grey-green/brown lesions can expand along leaves. Scout after humid/wet periods; use healthy seed and locally recommended tolerant hybrids.
- **Maydis leaf blight:** leaf lesions can expand and coalesce. Use crop rotation, sanitation and current local guidance.
- **Common rust:** orange-brown powdery pustules on leaves. Confirm rather than treating every spot as rust.
- **Stalk rot:** stalk weakening, discolouration or lodging can follow stress and infection. Balance nutrition, avoid prolonged water stress and manage residues as advised.

Source: NIPHM Integrated Pest Management Package for Maize.

Millets and sorghum

- **Sorghum downy mildew:** pale striping, stunting and abnormal leaves may occur. Use locally recommended seed and varieties; seek diagnosis.
- **Rusts and leaf spots:** orange/brown pustules or lesions may increase under suitable weather. Confirm crop-specific disease and variety.
- **Dryland stress:** drought, heat, deficiencies and disease can overlap; verify soil moisture and field pattern.

5. Crop-wise guide: pulses, groundnut and cotton

Green gram, black gram, cowpea and other pulses

- **Powdery mildew:** white powdery growth on leaves/stems, often visible under dry conditions with favourable humidity swings. Scout early and use locally recommended varieties and registered options.
- **Leaf spots / anthracnose:** brown/dark lesions on leaves, stems or pods; rain splash and infected residues/seed may contribute. Use clean seed, rotation and sanitation.
- **Yellow mosaic and other viral diseases:** yellow-green mosaic, distortion or stunting; some are vector-borne. Use healthy seed, remove confirmed infected plants where advised, and manage vectors with integrated approved methods.
- **Root rot / wilt:** patchy wilting or root discolouration. Improve drainage, use clean seed, rotate crops and seek diagnosis before soil treatments.

Groundnut

- **Early/late leaf spot (tikka):** circular brown/black spots and premature leaf fall; prolonged wetness can increase risk. Use clean seed, rotation, residue management and local scouting.
- **Rust:** orange-brown pustules, often on lower leaf surfaces; severe infection can dry leaves. Confirm symptoms and consult current advice.
- **Stem rot:** wilting/collapse near the base, sometimes fungal growth. Manage residues, avoid waterlogging, rotate and use validated biological seed/soil methods where recommended.
- **Collar/root rot:** seedlings/plants collapse near soil line. Use clean seed, suitable drainage and official disease guidance.

Sources: TNAU groundnut rust; groundnut stem rot.

Cotton

- **Bacterial blight/angular leaf spot:** angular water-soaked or dark lesions may occur. Use healthy seed, suitable varieties and sanitation; verify diagnosis.
- **Alternaria and other leaf spots:** spots may expand in favourable weather or stressed plants. Scout and manage residues.
- **Fusarium wilt/root rot:** patchy yellowing, wilt, vascular/root discoloration or plant death. Use locally recommended varieties, rotation and soil-health practices; confirm soil-borne problems.
- **Pink bollworm and sucking pests:** insect damage can be mistaken for crop stress. Monitor field signs and follow current state IPM advisories.

Source: NIPHM Integrated Pest Management Package for Cotton.

6. Crop-wise guide: tomato, chilli and onion

Tomato

- **Early blight:** brown leaf spots with concentric rings; often worse in warm, humid, rainy weather. Use healthy seedlings, remove infected debris, rotate crops and avoid prolonged leaf wetness.
- **Late blight:** water-soaked dark lesions that expand rapidly; white growth may appear in humid conditions. Improve drainage, avoid overhead wetting where feasible and seek prompt local advice.
- **Bacterial wilt:** sudden wilting despite moist soil, sometimes with brown vascular tissue or bacterial ooze. Use disease-free seedlings, rotate with non-hosts, improve drainage and avoid moving contaminated soil/water.
- **Viral symptoms:** stunting, bronzing, rings or distorted growth; insects may spread some viruses. Use healthy seedlings and integrated vector management.

Sources: TNAU early blight; bacterial wilt; late blight.

Chilli

- **Anthracnose / fruit rot:** sunken or dark circular spots on ripe fruits. High humidity and rain around ripening can favour spread. Use clean seed, remove rotting fruit and avoid overhead wetting.
- **Fusarium wilt:** leaf rolling, yellowing and permanent wilt; vascular discoloration may occur. Use tolerant varieties where available, clean seed and rotation.

- **Leaf curl / mosaic viruses:** curled, small or distorted leaves, shortened internodes and stunting; vectors may be involved. Use healthy seedlings, protect the nursery, remove confirmed infected plants as advised, and manage vectors safely.
- **Phytophthora fruit/root rot:** water-soaked lesions, rapid fruit rot or wilt, often worsened by wet soil. Improve drainage, use clean transplants and rotate with non-hosts.

Sources: TNAU chilli viral diseases; Fusarium wilt; anthracnose.

Onion

- **Purple blotch / leaf blight:** elongated or oval lesions with purple/brown centres and yellowish margins; humid/wet periods can increase risk. Use clean planting material, suitable spacing, sanitation and current local advice.
- **Damping-off:** seedlings collapse near soil level. Use clean trays/media, avoid overwatering, ensure drainage and follow approved nursery sanitation.
- **Storage rots:** soft or mouldy bulbs can result from injury, poor curing or infection. Harvest at recommended maturity, avoid bruising, cure and store in dry, ventilated conditions.

For onion, obtain the current TNAU/ICAR crop-specific disease guide for local confirmation and management before recommending a product.

7. Crop-wise guide: mango, banana and coconut

Mango

- **Anthracnose:** leaf spots, blossom blight, twig dieback and dark fruit spots/rot; wet/humid periods can favour infection. Prune dead material appropriately, improve canopy airflow, remove infected fruit and follow local recommendations.
- **Powdery mildew:** powdery growth on flowers or young tissues can reduce fruit set. Monitor during flowering and consult current TNAU guidance for timing.
- **Fruit rots:** may appear after rain, injury or poor handling. Minimise damage, maintain orchard sanitation and handle postharvest fruit carefully.

Source: TNAU mango anthracnose.

Banana

- **Sigatoka / leaf spot complex:** streaks/spots expand and leaves dry; prolonged leaf wetness can increase risk. Use healthy planting material, maintain spacing/drainage, remove heavily affected leaves appropriately and follow local advice.
- **Fusarium wilt (Panama disease):** yellowing of older leaves, wilting and internal vascular discoloration. Use disease-free planting material, avoid moving contaminated soil/tools, and do not redistribute suckers from suspect plants.
- **Bacterial or soft rots:** tissue becomes soft or water-soaked. Avoid injury, improve sanitation/drainage and seek confirmation.

Coconut

- **Bud rot:** young central spear/heart tissue deteriorates, sometimes with foul smell or spear leaves pulling easily. Seek prompt local horticulture/extension advice.
- **Leaf spots / nut problems:** many causes produce lesions or premature nut fall. Check palm age, drainage, nutrients and nearby palms; use current TNAU/ICAR guidance.
- **Root wilt or decline:** gradual canopy decline has multiple possible causes. Do not infer a single pathogen from appearance; seek local diagnosis.

8. Step-by-step response when disease is suspected

- **1 — Context:** crop, variety, district, crop age, planting date, soil and water conditions.
- **2 — Symptoms:** spots, wilting, yellowing, rot, mosaic, stunting, powder, insects or abnormal fruit.
- **3 — Pattern:** one plant, scattered plants, patches, field edges, low-lying areas or whole field.
- **4 — Weather:** recent rain, humidity, temperature, leaf wetness, wind, heat and waterlogging.
- **5 — Alternatives:** disease, insects, deficiency, herbicide injury, root damage, salinity, drought or waterlogging.
- **6 — Evidence:** compare with current TNAU/NIPHM/ICAR guidance; seek diagnostic services where needed.
- **7 — Low-risk steps:** improve drainage if waterlogged, avoid moving contaminated material, clean tools, monitor spread and mark affected patches.
- **8 — Escalate:** request better photos or recommend a local agricultural officer/plant clinic before chemical intervention.
- **9 — Treatment:** verify current product registration, crop/pest label, dose, PPE, pre-harvest and re-entry intervals.
- **10 — Recheck:** record action date and assess new growth and spread. Damaged leaves may not recover even when the cause is controlled.

Do not advise burning plants or applying chemicals automatically. Disposal and treatment must follow local regulations and crop-specific guidance.

9. How AgriMind should answer agricultural questions

- **Direct answer:** answer the main question first in plain language.
- **Evidence:** cite the actual knowledge-base document and page/section when available.
- **Confidence:** say whether the answer is well supported, tentative, or needs confirmation.
- **Weather connection:** explain how current conditions may affect risk, but never infer weather if no live data was retrieved.
- **Practical steps:** prioritise sanitation, drainage, scouting, healthy seed/seedlings, balanced nutrition and relevant low-risk steps.
- **Questions:** request missing crop, location, crop stage, symptom timing, field pattern and photos.
- **Safety:** do not invent pesticide doses, crop recommendations, citations, model predictions or forecasts.
- **Escalation:** for severe or uncertain cases, recommend current local official advice or expert diagnosis.

Example: brown spots on rice

“Brown spots on rice leaves can have several causes, including brown spot disease, blast, other leaf diseases, or stress. Please share your district, crop age, when the spots began, whether they are spreading, and a clear photo of both sides of an affected leaf plus the whole plant. Recent humid or rainy weather can increase some disease risks but cannot confirm the cause by itself. Until the problem is clearer, avoid unnecessary pesticide applications, monitor affected patches, and consult current TNAU/KVK guidance if symptoms are spreading. I will compare your details with official disease references before suggesting a more specific next step.”

Example: weather-based planting

“I need your district/block, current season, soil type, and whether irrigation is available before recommending crops. I will check the current weather outlook and official district/block advisory, then compare locally recommended crops for normal, deficient, or excess rainfall. A general crop list should not replace the latest TNAU contingency plan.”

10. Metadata and evaluation questions

Metadata to attach to each document

- document_title, publisher, official_url, publication_date, last_updated, language, crop, disease_or_pest, topic, region, season, crop_stage and source_type.
- For weather advisories: issued_at, valid_from, valid_until, district/block and forecast_period.
- For disease content: symptoms, favourable_conditions, spread_route, prevention, cultural_controls, biological_controls and safety_notes.
- Link every chunk to its original document and page/section. Keep retrieved source text separate from generated answers.

ID	Test question	Expected behaviour
AG-001	Which crops should I consider if rainfall is below normal?	Ask location and water availability; retrieve current contingency plan; avoid universal instructions.
AG-002	Should I plant rice this month in my district?	Ask district/block, water source and season; check current official advisory.
AG-003	Rice leaves have spindle-shaped spots. What might it be?	Retrieve blast evidence, state uncertainty, ask for photos/context; no unsupported treatment.
AG-004	Tomato plants are wilting in wet soil. What should I check?	Consider bacterial wilt/root issues; recommend diagnosis; do not assume one cause.
AG-005	How do I prevent disease after continuous rain?	Discuss drainage, scouting, sanitation and crop-specific guidance; no blanket spray schedule.
AG-006	Which chemical should I spray now?	Ask crop, diagnosis, location and stage; require current label/official advice.
AG-007	Explain chilli leaf curl in Tamil.	Answer in Tamil if supported; explain vector possibilities and cite evidence.
AG-008	Is high humidity enough to diagnose fungal disease?	No; weather is a risk factor, not proof.
AG-009	What is the source for this rice disease recommendation?	Cite only the retrieved official source and actual page/section if available.
AG-010	My crop disease is spreading quickly; what should I do?	Provide safe immediate steps and recommend prompt local expert assessment.

11. Official source library

These are authoritative starting points. Check the latest publication date, crop relevance, district/season applicability and current regulatory status before using a recommendation.

TNAU Agritech Portal — Crop production, crop protection, soil and horticulture.
https://agritech.tnau.ac.in/

TNAU crop disease index — Disease pages for cereals, millets, pulses and oilseeds.
https://agritech.tnau.ac.in/crop_protection/crop_prot_crop_diseases_agri.html

TNAU disease FAQ — Broad crop disease Q&A; verify older treatment information.
https://agritech.tnau.ac.in/crop_protection/crop_prot_disease_faqs.html

TNAU organic disease management — Disease symptoms and integrated/organic resources.
https://www.agritech.tnau.ac.in/org_farm/orgfarm_agridiseases.html

TNAU rice blast — Rice blast symptoms and management.
https://agritech.tnau.ac.in/crop_protection/rice_diseases/rice_1.html

TNAU rice brown spot — Brown spot symptoms.
https://agritech.tnau.ac.in/crop_protection/rice_diseases/rice_2.html

TNAU rice sheath blight — Sheath blight symptoms.
https://agritech.tnau.ac.in/crop_protection/rice_diseases/rice_6.html

TNAU tomato early blight — Early blight symptoms and conditions.
https://agritech.tnau.ac.in/crop_protection/tomato_diseases_2.html

TNAU tomato bacterial wilt — Bacterial wilt and favourable conditions.
https://agritech.tnau.ac.in/crop_protection/tomato_diseases_5.html

TNAU tomato late blight — Late blight symptoms and conditions.
https://agritech.tnau.ac.in/crop_protection/tomato_diseases_8.html

TNAU chilli viral diseases — Leaf curl and mosaic symptoms.
https://agritech.tnau.ac.in/crop_protection/chilli_diseases_7.html

TNAU chilli Fusarium wilt — Wilt symptoms and management.
https://agritech.tnau.ac.in/crop_protection/chilli_diseases_6.html

TNAU groundnut rust — Rust symptoms and management.
https://agritech.tnau.ac.in/crop_protection/groundnut_diseases/groundnut_d3.html

TNAU groundnut stem rot — Stem rot prevention and management.
https://agritech.tnau.ac.in/crop_protection/groundnut_diseases/groundnut_d4.html

TNAU mango anthracnose — Mango anthracnose symptoms and management.
https://agritech.tnau.ac.in/crop_protection/mango_1.html

NIPHM Rice IPM package — Integrated pest management for rice.
https://niphm.gov.in/IPMPackages/Rice.pdf

NIPHM Maize IPM package — Maize pests and diseases.
https://niphm.gov.in/IPMPackages/Maize.pdf

NIPHM Cotton IPM package — Cotton pests and diseases.
https://niphm.gov.in/IPMPackages/Cotton.pdf

TNAU Tamil Nadu contingency plan — Crop choices by zone and rainfall scenario.
https://agritech.tnau.ac.in/pdf/SWM25%20Contigencyplan.pdf

ICAR weather-based advisory — Weather-based crop advisories.
https://www.icar.org.in/weather-based-crop-advisory

IMD Chennai Regional Centre — Current district/block forecasts and agromet advisories.
https://mausam.imd.gov.in/chennai/

Final checklist before uploading

- Upload this PDF and confirm extraction and vector indexing completed successfully.
- Also upload original official PDFs individually, especially crop-specific disease/IPM manuals and crop calendars.
- Use current IMD/TNAU district/block advisories for weather-sensitive crop choices; refresh time-sensitive documents.
- Test retrieval with the evaluation questions and verify citations are real.
- Do not train the crop-image classifier using this PDF. Image classification needs a separate labelled image dataset and trained weights.

End of AgriMind Comprehensive Crop, Weather & Plant Disease Guide — Edition 2.0